

Introduction to Automation Testing and Selenium

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Agenda

Understand automation testing and Selenium basics to get started with automation.

- What is automation testing?
- The necessity of automation testing.
- Manual testing vs. automation testing
- When to automate software?
- Ways of automation testing.
 - Which one is suitable in which situation?
- Web UI automation
- Selenium
- Hands-On demo

What is automation testing?

Automation testing is the process of using specialized tools or scripts to execute test cases and validate the application's functionality, rather than performing tests manually.

Necessity of automation testing

- Efficiency and Speed
- Accuracy
- Cost-effectiveness
- Scalability
- Reusability
- Test Coverage
- Multi-environment Testing

Manual testing vs. Automation testing

Manual Testing	Automation Testing
Testing performed by humans without using tools or scripts.	Testing is done using tools or scripts to execute test cases automatically.
Slower due to human effort.	Faster as tests are automated.
Prone to human errors.	Highly accurate if scripts are correct.
Test cases need to be re-executed manually.	Test scripts can be reused multiple times.
Exploratory, usability, and ad-hoc testing.	Regression, performance, and repetitive testing.
Requires domain knowledge and minimal technical skills.	Requires programming and tool expertise.

Manual testing vs. automation testing

- Does automation testing replace manual testing?
- Have we automated all our test cases?

Manual testing vs. automation testing

- Why Automation Cannot Replace Manual Testing:

- Exploratory Testing
- Usability Testing
- Ad-hoc Testing
- Initial Stages
- Cost and Effort

- Factors Influencing Test Automation Coverage:

- Test Case Stability
- Feasibility
- Cost vs. Benefit
- Tool Limitations
- Human Insight

Product eligibility for automation testing

When Can We Decide a Software is Ready for Automation Testing?

- Stability of the Application
- Clear and Well-Defined Requirements
- Availability of Repetitive Test Cases
- Test Environment Readiness
- Feasibility of Automation
- Cost-Benefit Analysis
- Team Expertise and Resources
- Long-Term Testing Needs

Types of Automation Testing

- Functional Testing
- Regression Testing
- Unit Testing
- Performance Testing
- Sanity Testing
- Smoke Testing
- End-to-End Testing
- Cross-Browser Testing
- • Data-Driven Testing

Ways of Automation Testing

- API Automation
- UI Automation

Question:

- Which one is suitable in which situation?
 - Large data entry.
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Web UI Automation

Web UI automation involves automating the testing of user interface (UI) elements and interactions of a web application to ensure functionality, usability, and responsiveness across browsers and devices.

Key Components of Web UI Automation

- Web Elements:
 - Buttons, links, text fields, checkboxes, dropdowns, and other UI elements.
 - Automation scripts interact with these elements to simulate user actions.
 - Browsers:
 - The application under test (AUT) is rendered in browsers (e.g., Chrome, Firefox, Edge).
 - Crossbrowser compatibility is often tested.
 - Automation Tools:
 - Frameworks like Selenium, Cypress, Playwright, or Puppeteer.
 - Support scripting in languages such as Java, Python, JavaScript, etc.
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Challenges of Web UI Automation

- Dynamic and frequently changing UIs require script maintenance.
 - Initial setup and script development can be time-intensive.
 - Not suitable for exploratory or subjective testing like UX evaluation.
 - Few limitations of testing tools as well.
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What is Selenium?

Selenium is a toolset for automating web browsers. It allows testers to write scripts to simulate user interactions with web applications.

Key Components of Selenium

Selenium WebDriver

- A core component that provides APIs to automate web browser interactions.
- Interacts directly with browser instances to perform actions like clicking, typing, and navigating.

Selenium IDE (Integrated Development Environment)

- A browser extension for recording and playing back user interactions with a web application.
- Suitable for beginners or quick test prototyping.

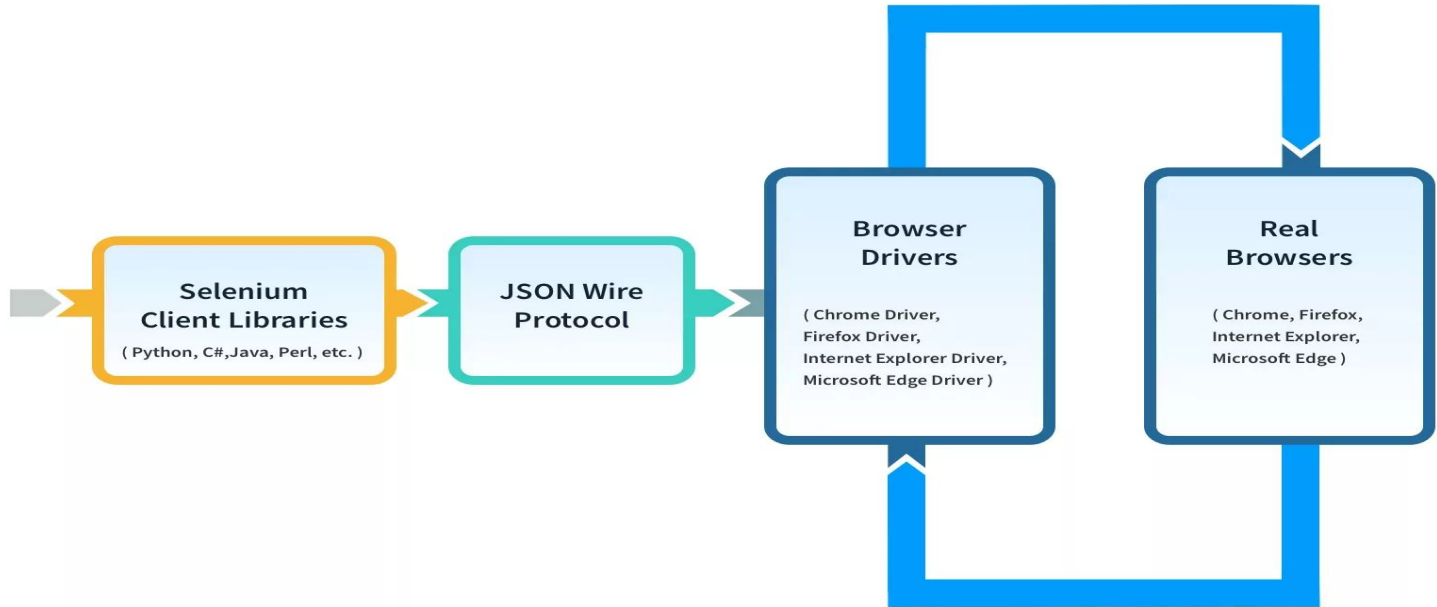
Selenium Grid

- Facilitates distributed and parallel execution of tests across multiple machines and browsers.
 - Ideal for large-scale cross-browser and cross-platform testing.
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Setting Up Selenium using Python

- Install Python
 - Install a code editor or IDE (e.g., VS Code, PyCharm)
 - Create and activate a virtual environment (optional but recommended)
 - Install Selenium using `pip install selenium`
 - Download the appropriate WebDriver (e.g., ChromeDriver, GeckoDriver)
 - Write and run your first Selenium script
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Selenium WebDriver Framework Architecture



Let's do a hands-on task to
start automation

Thank you!